MIL-C-80036C
19 December 1985
SUPERSEDING
MIL-C-80036B
25 September 1973

MILITARY SPECIFICATION

CUTTING MACHINE, OXYGEN FLAME AND PLASMA ARC, AUTOMATIC, STATIONARY, PHOTOELECTRIC TRACER CONTROLLED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers stationary, automatic, flame and plasma arc cutting machines for cutting various shapes from steel plate or other materials. The machines are controlled by a photoelectric tracer capable of tracing line drawings. The plasma arc system used with the cutting machines is covered by MIL-C-80090.

1.2 Classification. This specification covers cutting machines of the following types and sizes. The type and size to be furnished, shall be as specified (see 6.2.1).

Type I - For cutting with an oxygen-fuel gas torch having straight line and contour cutting speeds with a range of not greater than 2 inches per minute (IPM) to not less than 40 IPM.

Type II - For cutting with a plasma arc torch or oxygen-fuel gas torch having straight line cutting speed with a range of not greater than 50 IPM to not less than 240 IPM and contour cutting speed of not greater than 2 IPM to not less than 120 IPM. "

Size 1 - 48 in. width x 96 in. length cutting capacity.
Size 2 - 60 in. width x 120 in. length cutting capacity.
Size 3 - 72 in. width x 144 in. length cutting capacity,
Size 4 - 96 in. width x 192 in. length cutting capacity.
Size 5 - 120 in. width x 240 in. length cutting capacity.
Size 6 - 144 in. width x 288 in. length cutting capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3433
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies, and Accessories, Packaging of,

MIL-C-80090 - Cutting System, Metal, Plasma Arc.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government publications. The following other Government publication form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6,3).

3.2 Design. The machine shall be new and one of the manufacturer's current models, capable of cutting material in accordance with the requirements specified herein. The machine shall be all purpose, capable of performing intricate shape and straight line cutting operations using a coordinated drive system for the combination of transverse and longitudinal movement. The machine shall be capable of multiple torch operation. The design shall be such as to provide convenient, efficient, and safe operation. All structural components shall be designed to withstand maximum operating conditions while maintaining the accuracies specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U s Customary System of Units NJS) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, pressure gauges, and other similar devices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The cutting machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions, described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at point of installation (see 6.4).

3.2.9 Fuel. Unless otherwise specified, the machine shall be equipped for use with acetylene fuel gas. When specified (see 6.2.1), the cutting machine shall be equipped for use with the fuel gas specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

3.2.10 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.11 Interchangeability, All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders-or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer' s standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

3.4 Components.

3.4.1 Carriage assembly. The carriage assembly shall be constructed of metal with sufficient strength and rigidity to insure performance as specified herein. The carriage assembly shall have provisions for the tracing unit to traverse the entire tracing table area, and for one torch holder to simultaneously traverse the same dimensions within the cutting area. The rail and wheel contact surfaces shall have a finish that will assure vibration-free operation and performance as specified herein. Coordinated transverse and longitudinal movements shall be provided by individual drive motors, with provisions to lock the drive in transverse or longitudinal position for making straight cuts, parallel cuts, and cuts at 90 degrees to the main rails. Driving surfaces of the carriage shall not be exposed to the cutting operation. Movements of the carriage shall be positive, with backlash between drive and driven members limited to the extent necessary to assure cutting operations within the tolerances specified in 3.6. A suitable air blast or wiper shall be incorporated, if necessary, to keep rail surfaces clean. The gas and electrical distributing system shall be integral with the carriage or cross beam, and shall be mounted thereon. Unless otherwise specified, the machine shall be furnished for right hand operation. Right hand operation shall be with the torches to the right and the controls to the left of the operator when facing the control station. When specified (see 6.2.1), the machine shall be furnished for left hand operation. Left hand operation shall be with the torches to the left and the controls to the right of the operator when facing the control station.

3.4.2 Carriage rails. Steel carriage rails shall be of the manufacturer's standard design and shall be of a type which readily permit extension by adding sections. When specified (see 6.2.1), additional sections shall be furnished. Travel stops or limit switches which prevent carriage overtravel shall be provided.

3.4.3 Coordinate drive. The coordinate drive shall provide directional movements for the machine carriage. The carriage shall be powered by servo motors which are coordinated electronically through a directional control unit actuated by the tracer.

3.4.4 Cutting torch, oxygen-fuel gas. The oxygen-fuel gas cutting torch shall be of the valved, hose, machine type with a cylindrical barrel and rack, and shall be designed for use with oxygen and the specified gas (see 3.2.9 and 6.2.1). The end of the torch, housing the fuel and oxygen valves, shall be manufactured of bronze or brass. The torch head shall be constructed to receive cutting tips of various sizes. Unless otherwise specified (see 6.2.1), four oxygen cutting torches shall be furnished with the type I machine and three with the type II machine.

3.4.5 Torch ignitor. A remote controlled automatic igniter shall be provided for each torch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

3.4.6 Cutting system plasma arc. When specified (see 6.2.1), the type II machine shall be equipped with a plasma arc cutting system. The plasma arc system shalll conform to MIL-C-80090.

3.4.7 Motorized torch holder. Unless otherwise specified (see 6.2.1), the type I machine shall be equipped with four motorized torch holder assemblies designed for mounting cylindrical oxygen fuel cutting torches as specified in 3.4.4. Unless otherwise specified (see 6.2.1), the type II machine shall be equipped with three motorized torch holder assemblies designed for mounting cylindrical oxygen fuel cutting torches, and one holder assembly designed for mounting a plasma arc cutting torch conforming to MIL-C-80090. The construction of the holder assemblies shall provide for manually positioning the assemblies at any point along the carriage, and for removal of the assemblies from the carriage. Provisions shall be made for tilting the torch forward for high speed ripping and tilting sideways for beveling. The vertical travel of the torch shall be sufficient for cutting material 8 inches thick. The torch holder suspension system shall be of sufficient strength and rigidity to prevent binding or distortion in any manner, and to assure operations within the tolerances specified herein. When specified (see 6.2.1), special torch holder suspension systems shall be furnished as required.

3.4.8 Photoelectric tracer. The machine shall be equipped with a photoelectric cell scanning tracer. The tracer shall follow a pencil or ink line drawing on suitable drafting or heavy white paper, or a contrasting color template. The scanning head shall not contact the drawing or template in any manner. The width of the traced line shall be not less than 0.020 inch nor greater than 0.040 inch. Performance of the tracer shall not be affected by extraneous light normally encountered in shop areas. The tracing head shall be equipped with an automatic shut-off device controlled by the scanner. The device shall stop the machine and turn off the flow of gases to all torches and electric power to the plasma torch if the scanner should lose the line, A warning light shall be installed in a position visible to the operator to indicate when the machine has stopped. The tracer shall have electronic kerf compensation to allow for kerf correction to 0.250 inch. Means shall be provided for adjusting the height of the tracing head with respect to the tracing table.

3.4.9 Operator control panel, All controls for operation of the photoelectric tracer, ignitor, preheat, cutting oxygen, positioning of torches, feed control, individual torch control, and movement of carriages shall be provided in the control station. Means shall be provided to manually steer and remotely control the machine from the operator's panel. The control panel shall be centrally located for ease of operation, and positioned so the operator has an unobstructed view of the work area and tracing table at all times. The control station shall be sealed against the entrance of dust and moisture.

3.4.10 Gas control equipment. The oxygen and fuel gas regulators and their associated gauges shall be conveniently located for ease of reference

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

and adjustment. The equipment shall include regulation of preheat fuel, preheat oxygen, and the cutting oxygen. An individual solenoid valve to control the cutting oxygen shall be furnished for each torch. A system for controlling preheat and fast piercing starts shall be furnished. Flow of gases shall be automatically controlled by solenoid valves which shall immediately close when the master control switch is turned to the off position. Regulation and manifolding shall be of sufficient capacity to accommodate the specified number of torches. The manifold shall have capped outlets for future installation of additional torches in accordance with requirements listed in table I. All other associated gas control equipment, including hoses and fittings necessary for internal machine connection and operation, shall be included. All components shall be mounted and protected to withstand the operating conditions encountered when the machine is operated at maximum rated load and speed. When hose lengths exceed 7 feet, suitable means shall be provided for relieving the weight and preventing entanglement and chafing of the hoses to each torch.

3.4.11 Template table. An independent, moveable template table of dimensions necessary to meet the requirements in this specification shall be furnished with the size machine specified. When specified (see 6.2.1), additional sections of template table shall be furnished. The table shall move on a set of rails, with height and level adjustment provided. Means shall be provided to quickly and easily position the table in any desired position along its track.

3.4.12 Electrical System. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), electrical input power shall be 115 volt, single phase, 60 Hz, AC.

3.4.13 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. All motors not mounted in enclosed compartments shall be of the totally enclosed type. Motor horsepower and speed shall be adequate to meet maximum operating conditions.

3.4.14 Solid state components, Solid state design shall be used throughout for electronic components,

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacities of the machine shall be not less than those specified in table I for the machine size specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities.

Capacities	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6
Width of cut, inches (transverse)	48	60	72	96	120	144
Length of cut, inches (longitudinal)	96	120	144	192	240	288
Diameter-circle cut, inches (min.-max.)	3-48	3-60	3-72	3-96	3-120	3-144
Manifold-number of torch outlets	4	6	6	6	8	8

3.6 Performance. The machine and all components shall function properly to meet the requirements specified herein. Accuracy of torch travel with respect to the template pattern shall be + 1/64 inch. The machine shall be capable of negotiating 90 degree turns with not more than a 1/8 inch radius at speeds up to 40 IPM. Machines equipped with 4 oxygen cutting torches, shall be capable of cutting SAE 1020 or other low carbon steel 8 inches thick at a rate of not less than 3 IPM; 6 inches thick at a rate of not less than 5 IPM; and 1/4 inch thick at a rate of not less than 28 IPM. Straight line and contour cutting speed ranges for type I and type II machines shall be as specified (see 1.2).

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), the machine shall be furnished with all standard equipment normally furnished by the manufacturer.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.10 Marking of instruction plates, indicating plates, and nameplates. All words on instruction plates, indicatin plates, and nameplates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background,

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a trans-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

parent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the drive system. The meter shall be of the nonresetting type and shall have a range of O to 99,999 hours in increments of one hour. The meter shall be sealed-to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching the maximum accumulative hours, the meter readout shall automatically revert to zero and continue to totalize time.

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data shall be in the English language.

3.13 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection re-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

quirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a, First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4) .

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.5, the test in 4.6,1, and the inspection in 4,7. Failure of the item to pass the examination, test, and inspection shall be cause for rejection.

4.5 Examination. The machine and equipment shall be visually examined to determine compliance with all requirements of this specification.

4.6 Tests.

4.6.1 Operational test. The machine shall be properly connected to the electric power source, the oxygen supply, and the specified fuel gas supply, All components of the gas system shall be thoroughly checked for external leaks. Gas pressures shall be set in accordance with the supplier's recommendations. Acetylene pressure shall not exceed 15 PSI. All torches shall be ignited and adjusted to obtain a neutral flame cone pattern. Ignition and flame cycle of the torches shall be repeated 10 times per minute for a minimum of 3 minutes. During the performance tests, the oxygen fuel gas torches shall maintain satisfactory continuous operation without adjustments, blowbacks, or backfires when the oxygen pressure is increased or decreased 6 PSI gauge pressure from its pre-selected setting. The machine shall be operated for 30 minutes, during which time all components shall be tested through their full range without performing any cutting operations. The machine shall be tested to assure that, when the tracer loses the line it is following, the machine drive, the flow of gases to all torches, and the electric power to the plasma torch shall be automatically shut off.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

4.6.2 Performance tests.

4.6.2.1 Depth of cut test. A sample of SAE 1020 or other low carbon steel 6 inches thick, 10 inches wide, and long enough to accommodate three or four torches as applicable, shall be cut. The time required for the 10 inch cut shall be not greater than two minutes. The cut 'edge of the kerf shall be square and the slag on the bottom of the cut shall be easily removable.

4.6.2.2 Speed of cut test. A sample of SAE 1020 or other low carbon steel 1/4" thick, 28 inches wide, and long enough to accommodate the four torches shall be cut. The time required for the 28 inch cut shall be not greater than one minute. The cut edge of the kerf shall be square and the slag on the bottom of the cut shall be easily removable.

4.6.2.3 Irregular pattern test, The-manufacturer's standard test template, containing acute angles, 90-degree angles, sharp radii, and circles shall be placed on the tracing table and scanned. For the purpose of this test, one torch shall be fitted with a trace pen in lieu of the tip, and a piece of white drafting paper of the appropriate size shall be "securely placed in the work area. The scanner shall be locked on a line of the template, and a complete tracing shall be made in the clockwise direction; then a retracing shall be made of-the template in the counter-clockwise direction This test shall be repeated no less than six times. The cutting machine shall follow the template or accepted copies of the template within + 1/64 inch accuracy.

4.7 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level-A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The machine described in this specification will be used for cutting straight lines, circles, arcs, and various shapes from metals and other materials.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification,

b. Type and size required (see 1.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

c. First article if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify torch fuel gas, if other than acetylene (see 3.2.9 and 3.4.4). If natural gas specify line pressure.

h. Painting, if different (see 3.3.5).

i. Left hand operation, if required (see 3.4.1).

j. Additional carriage rail sections, if required (see 3.4.2).

k. Number of oxygen cutting torches required, if different (see 3.4.4).

l. Plasma arc cutting system, if required (see 3.4.6).

m. Motorized torch holder requirement, if different (see 3.4.7).

n. Type of torch holder suspension system, if different (see 3.4.7).

o. Additional sections of template table, if required (see 3.4.11).

p. Electrical system and input power supply, if different (see 3.4.12).

q. Size and capacity, if different (see 3.5).

r. Additional performance features, if required (see 3.6.1).

s. Standard equipment, if different (see 3.7),

t. Specify and fully describe optional equipment required (see 3.8).

u. Specify applicable repair parts, if required (see 3.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

w. Nameplate, if different (see 3.10.2).

x. Hour meter, if required (see 3.11).

y. First article inspection, if different (see 4.3).

z. Quality conformance inspection, if different (see 4.4).

aa. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, examination, and test reports, and other instructions for operation and maintenance should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern the machine in its operating environment such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual machines in the environment. The procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workplaces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.1 (f).

6.5 Cross reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

The numerical control requirements included in MIL-C-80036B have been deleted. The requirements for a numerical control cutting machine, oxygen flame and plasma arc, automatic, stationary are now covered in new specification MIL-C-80265.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 99

Review activities:
Air Force - 84
DLA -GS

User activities:
Navy - MC, SH

Preparing activity:
DLA-IP

(Project 3433-0092)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

ISTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in bmitting comments and suggestions for improvements. All users of military standardization documents are invited to provide ggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and iled. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the oblems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an knowledgement will be mailed to you within 30 days to let you know that your comments were received and are being nsidered.

OTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of ecification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

ENSE LOGISTICS AGENCY
mmander
fense Industrial Plant Equipment Center
TN: DIPEC-SSG
mphis, TN 38114-5297

FICIAL BUSINESS
NALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 4966 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Industrial Plant Equipment Center
ATTN: DIPEC-SSG
Memphis, TN 38114-5297

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-80036C	Cutting Machine Oxygen Flame and Plasma Arc, Automatic, Stationary, Photoelectric Tracer Controlled

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:33Z
Check the source to verify that this is the current version before use.